

AI SYSTEMS / QUALITY ENGINEERING / AUTOMATION

HOANG DO



Software Quality Engineer AI Systems and Automation Architect

Engineering AI that investigates software - and building the systems that test AI.

25+ YEARS

Software engineering, enterprise quality, automation, and technical problem solving.

ADOBE

Quality engineering across Marketo Engage, AJO B2B, email infrastructure, and cloud services.

AI BUILDER

Architect of InK-9 AI and the Echo testing, evaluation, and load platform family.

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Adobe

Senior Software Quality Engineer

San Jose, California

Builds automation and quality systems for enterprise marketing platforms, working across UI, APIs, backend services, data pipelines, email infrastructure, cloud environments, and production investigations.

SELECTED DELIVERY

- Delivered end-to-end validation for M1.5 remote-flow email infrastructure and cloud migration paths across legacy and AWS environments.
- Automated critical Marketo and AJO B2B workflows covering segmentation, nurture, A/B testing, personalization tokens, shared communication limits, and campaign execution.
- Validated email-platform capabilities including FES MTA behavior, DKIM checks, delivery workflows, and production sanity across multiple releases.
- Investigated AEP-to-Marketo data behavior, including UPS merge policy, datasets, profiles, segmentation, and cross-system integration issues.
- Built Python utilities and data simulators for faster setup, debugging, environment validation, and Databricks-backed test scenarios.
- Modernized UI coverage with Playwright and TypeScript while maintaining Selenium, Java, and WebDriverIO suites where appropriate.
- Used Splunk, Snowflake, S3, Databricks, Jenkins, APIs, and direct data inspection to turn complex failures into reproducible engineering evidence.
- Partnered with engineering, product, architecture, and release teams to validate releases and resolve problems spanning frontend, services, messaging, and customer data.

HOW I WORK

SYSTEM VIEW

Follow a customer workflow across UI, service, message, log, and data layers until the failure is explained.

AUTOMATE

Turn repeatable investigations and release checks into maintainable frameworks and utilities.

SHIP EVIDENCE

Give engineers concise reproduction steps, logs, queries, artifacts, and a clear confidence level.

AI that investigates, evaluates, and scales.

InK-9 AI

Multi-Agent Engineering Investigation Platform

Designed and built an internal platform that routes complex Marketo and platform issues to specialized agents, gathers evidence from engineering systems, and produces a structured verdict.

- Master orchestration for triage, routing, plan review, cross-agent assessment, and final synthesis.
- Specialist agents for AI/privacy jobs, conversational sessions, Marketo lead/email flows, and AEP data/profile investigations.
- Autonomous plan/action loop with confidence stopping, human escalation, and learned investigation patterns.
- Tiered model routing, session isolation, streaming UI, and Bedrock, Splunk, and Marketo integrations.

STACK FastAPI | JavaScript/HTML | SSE/WebSocket | AWS Bedrock | Splunk | Marketo APIs

Echo AI

End-to-End AI Testing

Autonomously tests Coworker AI through real browser conversations.

- Playwright drives sessions and captures chat API/SSE evidence.
- LLM decisions continue or complete each mission, followed by outcome verification.
- Detects runtime and authorization failures, creates bug artifacts, tracks tokens, and publishes Splunk telemetry.

PYTHON | PLAYWRIGHT | BEDROCK | SPLUNK

Echo Evals

LLM-as-Judge Evaluation

Evaluates response quality and detects regressions across completed Coworker AI sessions.

- Scores correctness, consistency, alignment, completeness, and helpfulness on a concrete 1-5 rubric.
- Per-session and cross-session analysis finds contradictions, broken runs, and weak spots.
- Flask dashboard, scheduled runs, and CLI workflows support Stage, QE, and Production.

PYTHON | CLAUDE | FLASK | JSON

Echo Loads

AI Load and Stress Testing

Measures reliability, latency, skill invocation, token use, and rate-limit behavior under parallel load.

- Jenkins orchestrates concurrent workers across Stage, QE, and controlled Production runs.
- Locust, Splunk, and HTML reports provide failure, latency, and per-skill analysis.
- Sample run: 564 requests and 564 skill invocations; 100% invoke rate, zero failures, and zero rate limits.

PYTHON | LOCUST | JENKINS | SPLUNK

Engineering range

Languages

TypeScript, JavaScript, Java, Python, C/C++, C#, J2EE, Node.js, AngularJS, ASP.NET, PHP and SQL; algorithms, data structures and design patterns

Testing and Automation

Playwright, Locust, Selenium, WebDriverIO, Cypress, QTP, TestNG, Cucumber, REST Assured, SoapUI and JMeter; Page Object and data-driven frameworks

Quality Engineering

Web, mobile iOS/Android and desktop; exploratory, black-box, white-box, API, UI, regression, performance, load and security testing

AI Cloud and Delivery

Agent orchestration, LLM evaluation, AWS, Jenkins, GitHub, CI/CD, Docker, Kubernetes, Jira, TFS and Trello

Data and Services

REST, SOAP, JSON, XML, SQL Server, MySQL, Oracle, MongoDB, Splunk, Snowflake, Databricks, S3, NumPy, Pandas and Matplotlib

Leadership and Methods

Agile, Scrum, Kanban, RUP, context-driven testing, project and risk management, product delivery, technical writing, training and mentoring

LEADERSHIP AND EARLIER EXPERIENCE

- More than 25 years across software engineering, quality engineering, release validation, production support, and enterprise web systems.
- Mentors engineers on automation design, investigation discipline, and maintainable test architecture; works effectively across global teams and time zones.
- Earlier in his career, helped build machine-learning products, established testing organizations, and published technical work on exploratory and context-driven testing.

INDEPENDENT PRODUCT WORK

Qunara | Founder and Product Builder

Building a bilingual platform for original editorial work and interactive reflection on quantum science, Buddhist philosophy, and human experience. Owns product vision, UX, development, and editorial direction.

POSITIONING

Software quality depth + AI-system architecture + enterprise product knowledge + hands-on delivery

Engineering foundation

eBay / AvenueCode

2021

Senior QA Automation Engineer

Owned API and UI automation for Purchase History, Buy It Now, and Add to Cart. Added 60 API and 30 UI tests, raised coverage from 70% to 85%, and sustained more than 90% pass reliability.

First Tech Credit Union

Feb 2020 - Feb 2021

Senior QA Automation Engineer

Built Java, Selenium, TestNG, and Maven automation for online banking. Automated more than 100 wire-transfer cases in 10 days and integrated execution into DevOps pipelines.

MeU Solutions

Sep 2014 - Dec 2019

Test Lead and Automation Lead

Led automation and exploratory testing for Forbes Travel Guide, Pragma ERP, and machine-learning test products. Built Java/Python frameworks, reduced manual testing cost by 30%, led a five-engineer team, and helped grow a new testing department beyond 40 engineers.

KMS Technology

Jul 2011 - Sep 2014

Software Tester and Test Lead

Led four automation engineers building a Java keyword-driven product and developed Page Object and hybrid frameworks. Defined test strategy, supported the testing center of excellence, led client teams up to ten engineers, and earned 4.5/5 customer satisfaction.

TRG-Enclave

Jul 2007 - Jul 2011

Senior QA Engineer

Led and grew testing teams, built Java Page Object automation, contributed C/C++ GIS development, and tested SAP-based ERP modules across material, warehouse, and HR management.

Global CyberSoft

Through 2007

Quality and Process Contributor

Served on the appraisal team that contributed to the organization achieving CMMI Level 4.

EDUCATION

B.S. Computer Science

HUFLIT University | 2001

CERTIFICATIONS

- PCAP Python Professional | 2021
- AWS Cloud Practitioner | 2020
- PMP | 2010
- PMI-RMP | 2012
- PMI-ACP | 2012
- Rapid Software Testing | 2016
- ISEB-ISTQB Certified Tester | 2010

PUBLICATIONS

Graph Builder for Exploratory Testing from a Novel Approach

SOMET 2018, Spain

A Novel Approach for Context-Driven Testing Using Heuristics and Machine Learning for Web Applications

INISCOM 2018, Da Nang | Springer